Noise Model-Based Line Segmentation for Plane Extraction in Sparse 3-D LiDAR Data

Linkun He, Bofeng Li, and Guang'e Chen

***Abstract*— Planar features serve as an important component in point cloud registration and reconstruction. However, extracting planes in the point clouds collected by a 3-D LiDAR sensor is still a challenging task due to the sparse property. To obtain reliable plane segmentation results, it is very necessary to fully exploit the scanning pattern of the sensor. In this article, we propose a novel plane extraction method for 3-D LiDAR data in a framework of point-to-line-to-plane. In the point-to-line stage, a new flat-point detector is introduced to obtain line segments. In the line-to-plane stage, we present the line based Douglas–Peucker (LBDP) algorithm to find coplanar line segments. Unlike region growing, which is generally applied to grouping line segments, LBDP does not suffer from the poor geometry of the selected initial region. More importantly, since the collected point clouds are always noisy, we model the measurement noise via statistical analysis, and bridge the noise level and parameter uncertainty to provide reasonable thresholds throughout our method. In the experiments, we evaluate the proposed method on both simulated and real datasets in terms of true positive rate (TPR), positive predictive value (PPV), $F1$ score, and five segmentation metrics. The results show that the proposed method can accurately extract planes in real time and outperforms the compared approaches.**

***Index Terms*— 3-D LiDAR, line segmentation, noise model, plane extraction.**

I. Introduction

THE laser scanning technology can efficiently capture numerous point patterns from object surfaces, recording the geometric and radiometric information of the real world. The obtained point clouds consist of discrete points without semantic information. As a consequence, the segmentation is indispensable for facilitating the interpretation of point clouds, by which the points are grouped into individual segments if they meet the merging criteria [1]. On the other hand, the point cloud segmentation is very helpful to remove redundant points, and provides a more compressive representation of the point cloud [2], [3].

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In human-made environments (e.g., indoor scenes and urban scenes), there are abundant objects with planar surfaces, such as floors, ceilings, walls, and building facades. Hence, planar patches are prevalent in the segmentation results, which have been proved to be robust geometric features for point cloud registration and reconstruction [4], [5], [6]. For these reasons, the extraction of planes (i.e., plane segmentation) has become a research hotpoint in the community of point cloud processing.

A. Plane Segmentation

Over last decades, a variety of plane segmentation approaches have been proposed, which can be basically categorized into four groups: model fitting based, region growing based, clustering based, and deep learning based.

1) Model Fitting-Based Method: Planes are simple geometric primitives that can be readily parameterized. Thereby, it is intuitive to use robust fitting methods to find planes in the point clouds. Random sample consensus (RANSAC) [7] and Hough transform (HT) [8] are the two most popular methods in this group [9]. The RANSAC paradigm mainly involves two steps: generating a candidate model and scoring, which are repeatedly conducted until the number of iterations reaches a certain threshold. The first step is implemented through randomly sampling a subset from the whole data. Since spatially close points are more likely to belong to the same plane, the predivision of the point cloud would help prune the sampling space and reduce spurious planes. Schnabel et al. [10] utilized the octree structure to establish spatial proximity among samples, and the sampling was restricted to a randomly selected cell from any level of the octree for each subset. Li et al. [11] split the point cloud space into a series of normal distribution transform (NDT) cells. To avoid spurious planes, all the cells were classified into planar and nonplanar cells. Then, one cell was randomly sampled among planar cells, and the points inside this cell were used to generate a candidate plane.

Different from RANSAC, HT is a voting-based algorithm which turns the plane segmentation task into a peak-finding problem in the accumulator. Borrmann et al. [12] reviewed several HT variants and presented a novel ball accumulator. Based on their evaluations, the randomized HT (RHT) [13] was considered as the method of choice when dealing with 3-D data. Grant et al. [14] proposed a fast RHT-based plane extraction method for point clouds collected by a rotating multilaser sensor (i.e., 3-D LiDAR sensor). Each scan line

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was divided into several groups by the detected breakpoints. Then, the voting was conducted among those groups, avoiding laboriously traversing all the points. Tian et al. [15] projected the point clouds onto the X-Y plane and used 2-D HT to extract straight lines corresponding to vertical planes in the 3-D space. Horizontal planes were extracted later via a similar strategy.

2) Region Growing-Based Method: As the name suggests, the region growing-based methods start with an initial region containing one or more points. The region is then iteratively expanded by adding the neighboring points that pass the coplanar tests. Generally, the vicinity is determined by the k-nearest neighbors (KNNs) or the fixed distance neighbors (FDNs) [16], [17], [18]. For organized point clouds (e.g., those captured by a time-of-fly camera), the image structure can be used to efficiently retrieve neighboring points [19], [20].

Given the large volume of point clouds, expanding the region by point could be time-consuming. A popular alternative is to use the octree which is extensively applied to point cloud segmentation [21], [22], [23]. In detail, the point cloud space is recursively subdivided into eight child nodes with the identical size until meeting a termination criterion. Those nodes presenting planar shape appearance are used as growth units. Moreover, two nodes are deemed neighbors if they have common vertices, edges, or faces. Wang and Tseng [24] stopped to further subdivide a node by checking the maximum distance to the best-fitting plane among all the points inside this node. Li et al. [25] generated the octree structure with adaptive thresholds obtained through statistics of node information.

Octree-based methods are appropriate for large-scale unorganized point clouds, showing promising efficiency and segmentation quality compared to point-based methods. Nonetheless, for organized point clouds, fully exploiting their distribution characteristics can further improve the efficiency and achieve competitive performance. Xiao et al. [26] proposed a fast subwindow based region growing method for plane segmentation in both structured and unstructured environments. Furthermore, with an insight into the scanning pattern of the corresponding laser sensor, scan lines can be partitioned into several line segments, which are grouped into planar regions by investigating their positions and orientations. It is easy to apply this idea to the point clouds that can be arranged into image structures [27], [28], [29], [30]. Besides, for mobile laser scanning (MLS) point clouds, the line segmentation can be conducted among scan profiles, i.e., the set of points captured by the scanner within each mirror revolution [31], [32].

3) Clustering-Based Method: By investigating the connectivity, distribution, density, and so forth, clustering techniques partition the points into groups, such that points in the same group have similar properties. Many classical clustering methods, such as K-means and DBSCAN, were modified and applied to plane segmentation [33], [34]. Besides, Trevor et al. [35] exploited connected components to cluster points into planes. There were two passes used: the first one for creating the label image and the second one for merging labels. Feng et al. [36] proposed the plane extraction using agglomerative hierarchical clustering (PEAC) algorithm. PEAC used nodes instead of points for clustering, avoiding the per-point normal estimation, and achieved more than 35 Hz frame rate in Kinect data. Proença and Gao [37] also employed the node initialization in PEAC. But the extraction objects were extended to planes and cylinders, which were applied to a visual odometry based on RGB-D camera. Another variant of PEAC was presented in [38]. The node size was dynamically adjusted and a more efficient clustering approach was introduced.

4) Deep Learning-Based Method: Nowadays, deep learning plays a nonnegligible role in environmental perception [39], [40], and plane segmentation is no exception. Some deep learning-based plane segmentation approaches were designed for specific planes, for example, the ground plane or building roofs [41], [42]. Considering a more general case, Li et al. [43] introduced a supervised neural network for fitting geometric primitives to point clouds, including planes, spheres, cylinders, and cones. This network can robustly detect a varying number of primitives at different scales without any user control. Lee and Jung [44] proposed the hybrid K-means plane segmentation (HKPS), where a neural network was trained for determining the K value for K-means.

Deep learning-based methods require a large amount of training data, leading to laborious labeling tasks. And their performance is expected to decrease when the environments are significantly different from those in the training data. Moreover, most learning-based methods present lack of interpretability, i.e., the so-called black box problem [45]. For the fields which set safety as the top priority, e.g., autonomous driving, this problem would hinder the application of deep learning-based methods.

B. Measurement Noise Model

During the plane extraction process, computing the plane parameters is always an inevitable task. For instance, in the scoring step of RANSAC, plane parameters are needed for ascertaining whether a point is an inlier. Properly modeling the measurement noise could help weight each point and produce more accurate plane parameters. Potential factors that influence the performance of a laser sensor are internal temperature, distance from the sensor, incidence angle, target texture, and so forth [46], [47], [48]. Generally, assuming the direction of the laser is accurately known, the statistics of the range noise is described as a Gaussian model with zero-mean and a standard deviation, and only geometric factors are considered in the literature. Kanazawa and Katanani [49] proposed renormalization to estimate plane parameters from ranging data, where the standard deviation was assumed to be linearly related to the distance from the sensor. According to [50], the range noise was proportional to the square of the range, and increased with the incidence angle. An analogous model was adopted in [51], but the impact of the incidence angle was neglected. Espadinha et al. [52] defined the standard deviation as a quadratic function of point depth and angle to the forward axis of the sensor.

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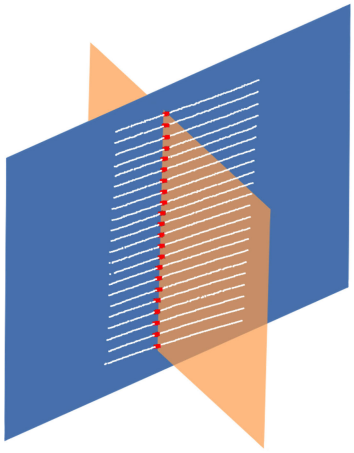


Fig. 1. Geometrically unreliable seeds constructed by three neighboring line segments (red). The wrongly estimated plane and the true plane are colored in orange and blue, respectively.

C. Our Contributions

Currently, with the development of simultaneous localization and mapping (SLAM) techniques, 3-D LiDAR sensors are extensively applied [53], [54]. Although many plane segmentation methods have achieved promising performance in dense point clouds, extracting planes from 3-D LiDAR data is still challenging.

1) To group discrete points into meaningful segments, the local reasoning serves as a key to plane segmentation. Due to the mechanical structure of the 3-D LiDAR sensor, the horizontal resolution is much higher than the vertical resolution, resulting in sparse and inhomogeneous point clouds. Therefore, the local reasoning methods prevailing in processing dense point clouds, such as voxelization and nearest neighbor searching, cannot function well on the 3-D LiDAR point clouds.
2) Traditional plane segmentation methods pay less attention to processing efficiency. However, 3-D LiDAR sensors are usually adopted for real-time applications, so the algorithm efficiency is of vital importance.
3) Since the 3-D LiDAR data is noisy, relying on empirical thresholds would cause unreliable results. To provide reasonable thresholds, it is very necessary to take the noise into consideration. Although some methods noticed this point, none of them established the correlation between the parameter uncertainty and the noise level, and the thresholds still lacked mathematical reasoning.

To overcome above challenges, this article presents a fast plane segmentation method for sparse 3-D LiDAR data, which has three main steps: 1) line segmentation; 2) line-based Douglas–Peucker (LBDP) algorithm, a robust approach for grouping line segments into planes; and 3) refinement. The proposed method fully exploits the scanning pattern of the 3-D LiDAR sensor, which provides us a perspective on local reasoning and allows this method to process each frame in real time. Moreover, a key contribution in this article is that we bridge the parameter uncertainty and the noise model via the error propagation theory. Benefitting from this, the proposed method can accurately extract planes in noisy point clouds with reasonable thresholds.

Some existing methods also exploited line segments [27], [28], [29], [30], but the proposed method significantly differs from them. First, considering the mechanical structure of 3-D LiDAR sensors, each laser forms a cone in space while rotating, and the intersection with a plane forms a conic. Hence, we do not conduct line segmentation along the scan line to avoid unnecessary systematic error. Second, previous methods usually use region growing to group line segments into planes, and several (basically two or three) neighboring line segments are selected as the initial region. However, these neighboring lines are spatially close to each other and would result in poor geometry (Fig. 1). With unreliable seeds, region growing is prone to produce fragmentary results. In contrast, LBDP does not start from a limited region. The planes are recursively extracted from a line sequence.

II. Method

The proposed plane extraction method consists of three primary steps (Fig. 2). First, we group flat points to obtain the line segments. Planes are later constructed by coplanar line segments which are found using the LBDP algorithm. Finally, we employ region growing and merging to cope with unassigned points and occlusion.

A. Noise Model Construction and Application

We first introduce the measurement noise model of the laser sensor, which plays a fundamental role in our method. For 3-D LiDAR data, we can trivially project the point cloud onto a range image according to the azimuth and elevation of each point. Let $\boldsymbol{p}_c^r = \rho_c^r \boldsymbol{m}_c^r$ represent one measurement in the point cloud, where r and c denote the row index and the column index in the range image, respectively. Each measurement can be expressed as the product of range ρ_c^r and measurement direction $\boldsymbol{m}_c^r$. We make the same assumption as that in [50] that $\boldsymbol{m}_c^r$ is accurately known while ρ_c^r is contaminated by Gaussian noise. Normally, the range ρ_c^r and the incidence angle γ_c^r are considered when formulating the noise. Since the effect of these two factors are not independent, we intend to establish a model considering only one of them. As it is difficult to find a simple model suitable for characterizing the range effect, we neglect the influence of the range on the measurement noise, i.e., $\rho_c^r \sim \mathcal{N}(\tilde{\rho}_c^r, \sigma^2(\gamma_c^r))$, where $\tilde{\rho}_c^r$ is the true range, and $(\cdot)$ encloses function arguments. Although only considering incidence factor may misassign distant points in large-scale environments, those points would not be utilized directly since they suffer from poor ranging accuracy and limited capability of capturing the geometry features.

A series of experiments were carried out in an underground garage to model the statistics of the measurement noise. We set the 3-D LiDAR sensor in front of a planar wall and collected point clouds with different sensor-to-wall distances. After that, we manually extracted the wall in each point cloud, and estimated the plane parameters using the least squares fitting, which are considered as noiseless. Note that some unreliable

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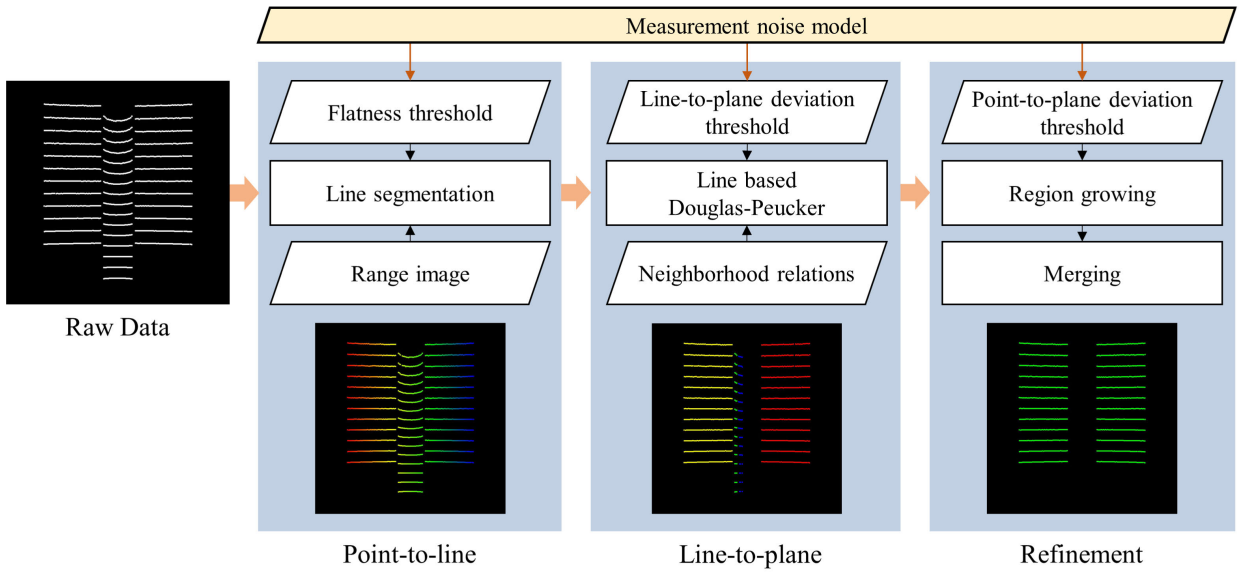


Fig. 2. Workflow of the proposed method.

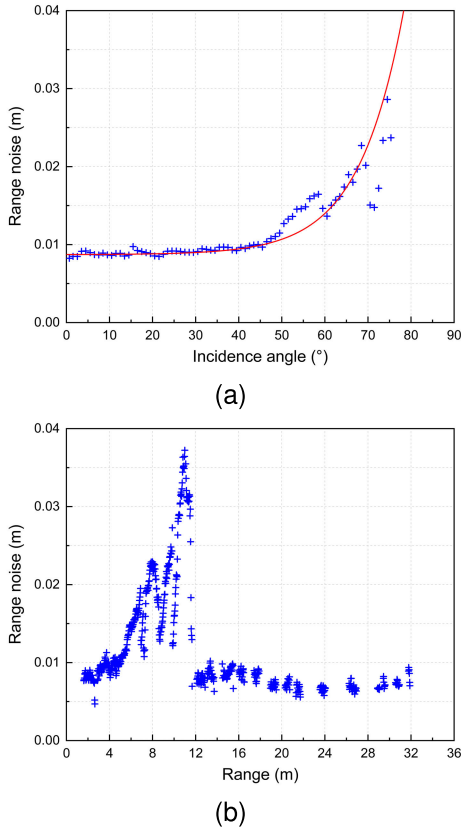


Fig. 3. (a) Angle-noise scatter plot with an interval of 1°, the red curve denotes the fit model. (b) Range-noise scatter plot with an interval of 0.03 m.

measurements may be found in the point clouds, in the cases that the laser beam scans the object edges or is parallel to the surface [55], [56]. These measurements obviously deviate from the real surfaces, presenting as outliers (e.g., the trailing phenomenon). We deem that these measurements are contaminated by gross errors, and therefore did not involve them in the noise model construction. With the plane parameters and $\boldsymbol{m}_c^r$ of each point, we can calculate γ_c^r as well as the range error. In our experiments, the error ranged from 8.75×10^{-10} to 0.18 m. Then, measurements were assigned to different sets according to the incidence angle with a fixed interval. Within each set, the mean of the incidence angles and the standard deviation of range errors were computed. The corresponding scatter plot is illustrated in Fig. 3(a). With the incidence angle increasing from 0° to 90°, the range noise initially increases slowly, and then increases rapidly when the incidence angle is larger than about 45°. Such a trend can be appropriately described by a sigmoid-like function

$$\sigma\left(\gamma_c^r\right) = \frac{1}{1+\exp\left(k_1\gamma_c^r + k_2\right)} + k_3 \tag{1}$$

where $\gamma_c^r \in [0, 90°)$, parameters k_1, k_2, and k_3 are gained by the least squares fitting. In comparison, the relationship between range noise and range is not clear [Fig. 3(b)].

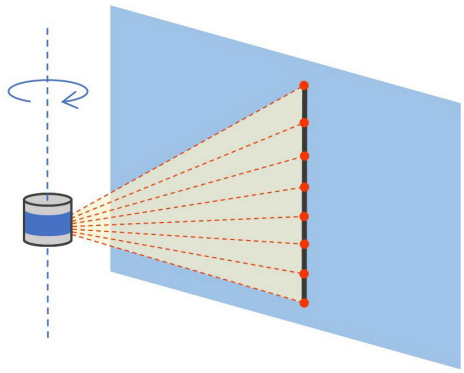


Fig. 4. Distribution characteristics of 3-D LiDAR data.

The purpose of noise model construction is twofold: we exploit the noise model to provide reasonable thresholds for identifying whether three points are collinear and whether a point is coplanar with a plane, which are served as keys to the point-to-line and line-to-plane stages, respectively.

1) Noise Model-Based Line Segmentation: After projection, the measurements on each column share the same azimuth under the LiDAR frame, so they are ideally distributed on a plane passing through the LiDAR origin (Fig. 4). As two

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nonparallel planes always intersect along a line, we can detect lines among points with the identical column index to identify the points on planar surfaces.

Sorting the points in one column by their row indices, consecutive flat points are considered to form a line segment. The flatness of one point is designed to be composed of range measurements, such that the uncertainty is easy to access. Given three consecutive points $\boldsymbol{p}_c^{r-1}$, $\boldsymbol{p}_c^r$, and $\boldsymbol{p}_c^{r+1}$, let $\boldsymbol{b}$ denote the angle bisector of the triangle formed by the LiDAR origin $\boldsymbol{O}$, $\boldsymbol{p}_c^{r-1}$, and $\boldsymbol{p}_c^{r+1}$. As the vertical resolution of the 3-D LiDAR sensor is always invariant, ρ_c^r should be equal to the length of $\boldsymbol{b}$ if $\boldsymbol{p}_c^{r-1}$, $\boldsymbol{p}_c^r$, and $\boldsymbol{p}_c^{r+1}$ are collinear. Thereby, we define the flatness of a point $\boldsymbol{p}_c^r$ as

$$f_c^r = \|\boldsymbol{b}\| - \rho_c^r = \frac{2\rho_c^{r-1}\rho_c^{r+1}\cos\theta_v}{\rho_c^{r-1}+\rho_c^{r+1}} - \rho_c^r \tag{2}$$

where θ_v is the vertical resolution of the sensor. Due to the existence of noise, a threshold is needed to distinguish flat points and nonflat points. Assuming measurements are independent, the uncertainty of f_c^r is estimated using the error propagation theory

$$\sigma_f^2 = \left(\frac{\partial f_c^r}{\partial \rho_c^{r-1}}\right)^2 \sigma^2\left(\gamma_c^{r-1}\right) + \left(\frac{\partial f_c^r}{\partial \rho_c^r}\right)^2 \sigma^2\left(\gamma_c^r\right) + \left(\frac{\partial f_c^r}{\partial \rho_c^{r+1}}\right)^2 \sigma^2\left(\gamma_c^{r+1}\right) \tag{3}$$

where

$$\frac{\partial f_c^r}{\partial \rho_c^{r-1}} = 2\cos\theta_v\left(\frac{\rho_c^{r+1}}{\rho_c^{r-1}+\rho_c^{r+1}}\right)^2 \tag{4}$$

$$\frac{\partial f_c^r}{\partial \rho_c^r} = -1 \tag{5}$$

$$\frac{\partial f_c^r}{\partial \rho_c^{r+1}} = 2\cos\theta_v\left(\frac{\rho_c^{r-1}}{\rho_c^{r-1}+\rho_c^{r+1}}\right)^2. \tag{6}$$

If $|f_c^r|/\sigma_f \geq T_\sigma$, $\boldsymbol{p}_c^r$ will be labeled as a nonflat point otherwise a flat point. It is noteworthy that the incidence angle is required for computing the variance of range noise, but plane information is unavailable at this stage. For simplicity and efficiency, we estimate the local normal of each point as the cross product of the vectors between the horizontal and vertical neighbors [57]. However, our plane extraction method is not limited to this algorithm, and can benefit from novel normal estimation approaches.

Readers may notice that (2) cannot be applied when $\boldsymbol{p}_c^{r-1}$ or $\boldsymbol{p}_c^{r+1}$ are missing. Moreover, point dropout occasionally happens during data acquisition, resulting in invalid pixels in the range image. To handle this issue, we would like to present a general version for computing the flatness. Given three consecutive valid points $\boldsymbol{p}_c^{r-m}$, $\boldsymbol{p}_c^r$, and $\boldsymbol{p}_c^{r+n}$ in one column, where $m, n \geq 1$, there are three potential cases.

1) $m = n$: In this case, $\boldsymbol{p}_c^r$ and the LiDAR origin still establish the angle bisector, so we only need to replace θ_v in (2) by $m\theta_v$ or $n\theta_v$.
2) $m > n$: Since the angle between $\boldsymbol{p}_c^{r-m}$ and $\boldsymbol{p}_c^r$ is larger than that between $\boldsymbol{p}_c^r$ and $\boldsymbol{p}_c^{r+n}$, we construct a virtual range $\hat{\rho}_c^{r-n}$ by interpolation. Using some simple geometry, we get

$$\hat{\rho}_c^{r-n} = \frac{\sin\alpha}{\sin(\alpha+(m-n)\theta_v)}\rho_c^{r-m}. \tag{7}$$

The angle α is illustrated in Fig. 5, which can be easily computed with the coordinates of $\boldsymbol{p}_c^{r-m}$ and $\boldsymbol{p}_c^{r+n}$. Then, (2) is rewritten as

$$f_c^r = \frac{2\hat{\rho}_c^{r-n}\rho_c^{r+n}\cos n\theta_v}{\hat{\rho}_c^{r-n}+\rho_c^{r+n}} - \rho_c^r \tag{8}$$

3) $m < n$: This is similar to the previous case, but the interpolation is conducted between $\boldsymbol{p}_c^r$ and $\boldsymbol{p}_c^{r+n}$.

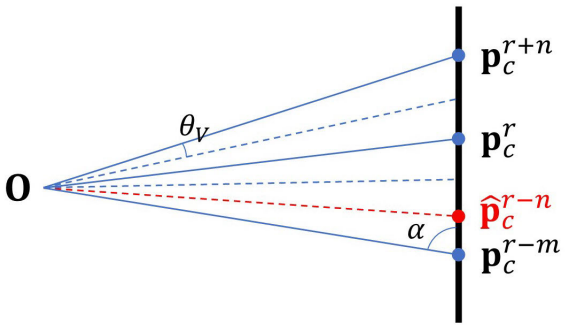


Fig. 5. Example of point dropout. Dashed lines represent missing data.

After labeling all points in the current column, consecutive flat points are grouped into the same segment. Besides, nonflat points are assigned to one of the neighboring segments. We compare the distances between a nonflat point $\boldsymbol{p}_c^r$ and its neighboring points $\boldsymbol{p}_c^{r-m}$, $\boldsymbol{p}_c^{r+n}$, and then $\boldsymbol{p}_c^r$ is assigned to the segment that the closer neighbor belongs to. Considering that very short segments are generally not reliable, those segments containing fewer than three points are discarded.

2) Noise Model-Based Coplanarity Identification: In the proposed method, we would repeatedly ascertain whether a point is coplanar with a plane, where the noise model is also employed. Herein, a plane is a set of coplanar points in nature, and we use the Hessian normal form for the plane representation, i.e., $\boldsymbol{n}\cdot\boldsymbol{p}_c^r = d$, where $\boldsymbol{n}$ is the unit plane normal, and d is the distance from the sensor. The solution of approximate least squares problem (ALSP) provided in [50] is adopted to efficiently estimate the plane parameters along with the 4×4 covariance matrix. Given a plane $\mathcal{P}$, the plane parameters are obtained by maximizing the following function:

$$\max_{\boldsymbol{n},d} \mathcal{L}_{\text{ALSP}} = -\frac{1}{2}\sum_{\boldsymbol{p}_c^r\in\mathcal{P}} \frac{\left(\boldsymbol{n}\cdot\boldsymbol{p}_c^r - d\right)^2}{\sigma^2\left(\gamma_c^r\right)}. \tag{9}$$

Let $\boldsymbol{H}$ denote the Hessian of (9), the corresponding covariance matrix $\boldsymbol{C}$ is estimated as the negative of the Moore–Penrose generalized inverse of $\boldsymbol{H}$.

Generally, the coplanar constraint is based on the point-to-plane distance. We further take the noise model and the uncertainty of plane parameters into consideration, and define the deviation of a point $\boldsymbol{p}_c^r$ from a plane $\mathcal{P}$ as

$$\text{dev}\left(\boldsymbol{p}_c^r, \mathcal{P}\right) = \frac{|\boldsymbol{n}\cdot\boldsymbol{p}-d|}{\sqrt{\boldsymbol{n}^{\mathrm{T}}\boldsymbol{C}_p\boldsymbol{n}+\boldsymbol{x}_c^{r\mathrm{T}}\boldsymbol{C}\boldsymbol{x}_c^r}} \tag{10}$$

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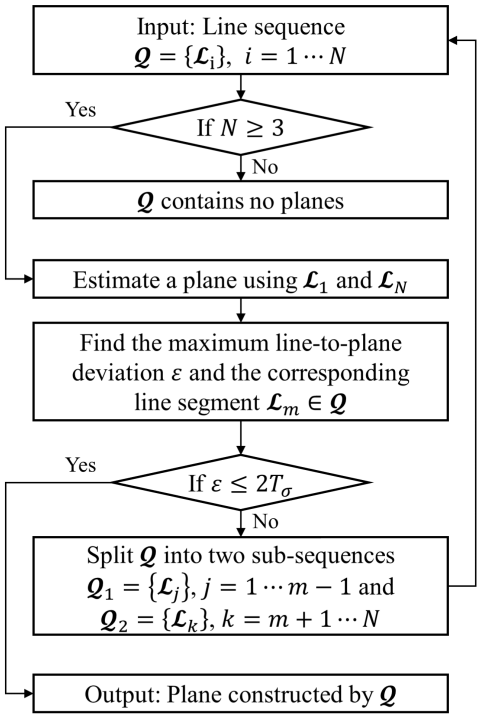


Fig. 6. Flowchart of the LBDP algorithm.

where $\boldsymbol{C}_p = \sigma^2(\gamma_c^r)\boldsymbol{m}_c^r\boldsymbol{m}_c^{r\mathrm{T}}$ is the covariance of $\boldsymbol{p}_c^r$ and $\boldsymbol{x}_c^r = [\boldsymbol{p}_c^{r\mathrm{T}}1]^{\mathrm{T}}$ is the homogeneous form of $\boldsymbol{p}_c^r$. If the deviation is less than T_σ, $\boldsymbol{p}_c^r$ will be deemed as coplanar with $\mathcal{P}$.

B. LBDP Algorithm

The Douglas–Peucker algorithm is a classical method frequently applied to simplifying polylines [58]. Here, we use a similar scheme to extract planes from line segments, namely the LBDP algorithm. To this end, we first need to group line segments into line sequences, where the neighborhood relations among line segments are required. For a line segment $\mathcal{L}_1 = \{\boldsymbol{p}_{c_1}^{r_1+i}\},\ i = 1, \ldots, N_1$, one left neighbor in column $c_1 - 1$ and one right neighbor in column $c_1 + 1$ are expected. In each column of interest, we deem the line segment having the maximum overlap with $\mathcal{L}_1$ a candidate. The overlap between $\mathcal{L}_1$ and a candidate neighbor $\mathcal{L}_2 = \{p_{c_2}^{r_2+j}\},\ j = 1, \ldots, N_2$ is quantified by

$$l = \min(r_1 + N_1, r_2 + N_2) - \max(r_1 + 1, r_2 + 1) + 1. \quad (11)$$

Note that $\mathcal{L}_1$ and $\mathcal{L}_2$ do not overlap when $l \leq 0$. $\mathcal{L}_2$ is further assessed using following constraints.

1) *Continuity:* Occlusion would result in a large gap between $\mathcal{L}_1$ and $\mathcal{L}_2$. In this case, they form a spurious plane that is almost parallel to the laser beam. Hence, we employ the angle β proposed in [20] to avoid neighbors from unconnected surfaces. For each pair of pixels $\boldsymbol{p}_{c_1}^r$, $\boldsymbol{p}_{c_2}^r$ in the overlapping area of $\mathcal{L}_1$ and $\mathcal{L}_2$, suppose $\|\boldsymbol{p}_{c_1}^r\| > \|\boldsymbol{p}_{c_2}^r\|$, β is calculated by

$$\beta = \arctan \frac{\|\boldsymbol{p}_{c_2}^r\| \sin\theta_h}{\|\boldsymbol{p}_{c_1}^r\| - \|\boldsymbol{p}_{c_2}^r\| \cos\theta_h} \quad (12)$$

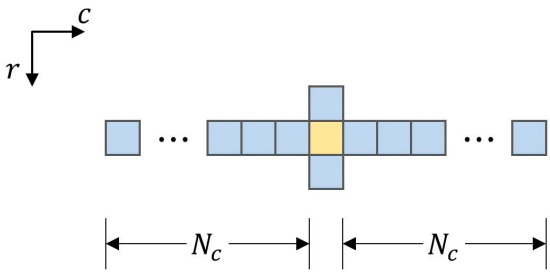


Fig. 7. Connected components (blue) of a point (yellow) in the range image.

where θ_h is the horizontal resolution of the sensor. If all pairs satisfy $\beta < T_\alpha$, $\mathcal{L}_2$ will not be regarded as a neighbor.

2) *Coplanarity:* Owing to the high resolution of 3-D LiDAR sensors in the horizontal direction, neighboring line segments will not sharply vary in direction if they are on the identical planar surface, that is,

$$\arccos|\mathrm{dir}(\mathcal{L}_1) \cdot \mathrm{dir}(\mathcal{L}_2)| < T_\alpha \quad (13)$$

where $\mathrm{dir}(\cdot)$ returns the normalized direction vector of the input line segment. For efficiency, we approximate the direction of a line segment as the vector established by two end points.

A line sequence $\mathcal{Q} = \{\mathcal{L}_i\},\ i = 1, \ldots, N$ is built by grouping the neighboring line segments in a region growing manner. The ungrouped line segment containing more points enjoys the priority to be selected as a seed. Different from existing methods, we use region growing to obtain line sequences instead of planes, where the plane parameters are not required. And this is the reason why LBDP is not influenced by the poor geometry of the seed. Note that since each line segment has at most one neighbor on each side, the line sequence can be sorted by column index with no effort. $\mathcal{Q}$ is then recursively subdivided into two subsequences until each subsequence contains only coplanar line segments (Fig. 6). The coplanar property of a line sequence is tested by the line-to-plane deviation. Here, the plane is formed by two "end lines" of the current line sequence. For a line segment $\mathcal{L}_i$, the line-to-plane deviation is defined as the sum of point-to-plane deviations (see Section II-A), and only two end points of $\mathcal{L}_i$ are involved in the calculation for the sake of efficiency. On that basis, a subsequence will not be further subdivided if the deviations of all member lines are below $2T_\sigma$. Besides, only line sequences containing more than two line segments can be subdivided.

Spurious planes could be found in the results of LBDP. This is because of the undesirable line segments from nonplanar surfaces, such as rough surfaces and curved surfaces that are locally planar. Nevertheless, we can investigate the scatter matrices to remove the spurious planes. Suppose $e_1 \leq e_2 \leq e_3$ are the eigenvalues of the scatter matrix, the value of e_1/e_2 reflects the planarity of the corresponding point set. Consequently, the planes with e_1/e_2 larger than T_e are filtered out. All the line segments that are not involved in any planes are degraded to discrete points.

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TABLE I
TECHNICAL PARAMETERS OF THE SENSORS

Sensor	Resolution	Field of View		Range
		Hor.	Ver.	
VLP-16	16×1800	360°	-15° ∼15°	100 m
HDL-32E	32×1800	360°	-30.67° ∼10.67°	100 m
OS1-64	64×1024	360°	-16.6° ∼16.6°	120 m

C. Incomplete Plane Refinement

After the line-level process, there are still discrete points that are potentially located on planar surfaces but are not assigned. To achieve higher completeness, we refine the results by point based region growing. The planes obtained in Section II-B are used as initial regions, and the range image is employed to efficiently retrieve neighboring points. An unassigned point will be added to the current region if the coplanar constraint is satisfied (see Section II-A).

Another aspect contributing to incomplete results is oversegmentation. In LBDP, the line segments from a complex-shaped plane can be assigned to several line sequences, and it is necessary to integrate the resulting planar patches into one plane. Besides, occlusion can also lead to severe oversegmentation. Therefore, a fitting error based merging algorithm is introduced. Given a pair of planes $\mathcal{P}_s$ and $\mathcal{P}_t$, let ε_s, ε_t, and ε_m denote the fitting errors of $\mathcal{P}_s$, $\mathcal{P}_t$ and the merged plane (i.e., $\mathcal{P}_s \bigcup \mathcal{P}_t$), respectively. If $\varepsilon_s + \varepsilon_t > \varepsilon_m$, $\mathcal{P}_s$ and $\mathcal{P}_t$ will be merged. To improve the efficiency and suppress undersegmentation, only neighboring planes with analogous normal vectors are considered for merging. Specifically, if a point and one of its connected components belong to different planes, the corresponding planes will be considered as neighbors (Fig. 7). The size of the connected components depends upon $N_c = T_\alpha/\theta_h$. Finally, the planes that cannot be merged with any other planes are defined as isolated planes. We discard those isolated planes that are linear (i.e., $e_2/e_3 \leq T_e$), because they potentially come from a cylindrical surface, which can be approximated as a series of narrow planes.

III. EXPERIMENTS

In this section, we evaluate the proposed plane extraction method in various scenes including simulated and real datasets. Using the simulated dataset enjoys the merit that we can add artificial noise to the measurements as we have assumed, so it is suitable for verifying the feasibility of our method. The real dataset further tests the robustness and applicability of the proposed method and demonstrate its capability of online processing in practice.

A. Simulated Dataset

We simulated 150 point clouds based on the model of the Velodyne VLP-16 LiDAR sensor (Table I). The parameters of the noise model we used were obtained by prior calibration: $k_1 = -9.8737 \times 10^{-2}$, $k_2 = 11.1626$, $k_3 = 8.7105 \times 10^{-3}$. The simulated dataset is composed of three subdatasets with the same size: 1) vertical; 2) horizontal; and 3) occluded. Each point cloud in the vertical dataset and the horizontal dataset contains two intersecting 5 × 5 m planes, and the dihedral angle varies between 80° and 170°. The intersection lines are parallel with the z-axis in the horizontal dataset, whereas perpendicular in the vertical dataset. The point clouds in the occluded dataset contain one 5 × 5 m plane and one cylinder in front of the plane, whose diameter ranges from 0.1 to 1.0 m. Apart from differences in shapes, the sensor-to-plane distance increases from 8 to 12 m, and the maximum and the minimum of the point density are 100.80 and 33.68 points/m^2, respectively.

These three subdatasets are designed to test each step of the proposed method. The performance on the vertical dataset relies on the line segmentation, while LBDP is expected to properly distinguish two planes in the horizontal dataset. The occluded dataset is used to reveal whether the proposed method can recover the complete planes from occlusion. Moreover, a series of experiments are conducted on the simulated dataset to search for the appropriate parameters. For quantitative evaluation, we use positive predictive value (PPV), true positive rate (TPR), and $F1$ score as metrics that indicate the consistency with the ground truth, and their definitions are as follows:

$$\text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{14}$$

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{15}$$

$$F1 = 2 \cdot \frac{\text{PPV} \cdot \text{TPR}}{\text{PPV} + \text{TPR}}. \tag{16}$$

If a detected point is assigned to the correct plane, it will be counted as true positive (TP), otherwise false positive (FP). Besides, FN stands for false negative, namely miss detection.

There are three parameters in the proposed method: 1) T_σ is the key parameter that determines whether intersecting planes can be clearly distinguished; 2) T_e contributes to the removal of nonplanar surfaces; and 3) T_α is the tolerance for considering two vectors have the same direction. During the experiments, T_σ is set to 1.0, 2.0, 3.0, and 4.0. T_e takes 0.01, 0.04, and 0.09, and T_α takes 5°, 10°, and 15°. The evaluation results show that the PPV keeps greater than 90% and is slightly affected by the changes in parameter setting (Fig. 8). However, TPR and $F1$ are apparently lower when T_σ is set to 1.0. Increasing T_σ from 1.0 to 2.0 improves the performance by a large margin. Nevertheless, continuing to increase T_σ shows a little improvement but lead to a higher risk of undersegmentation, particularly from 3.0 to 4.0. Also, $T_\alpha = 5°$ is too tight for this method, resulting in relatively poor performances. Choosing $T_\alpha = 10°$ or $T_\alpha = 15°$ achieves similar performances, but a smaller value can facilitate the efficiency. For example, the frequency of calculating the fitting error will decrease during merging. As for T_e, $T_e = 0.09$ produces some spurious planes, which imposes a weak constraint on the planarity. On the other hand, $T_e = 0.01$ tends to miss planes with limited widths due to noise. Therefore, a moderate value is preferred to balance

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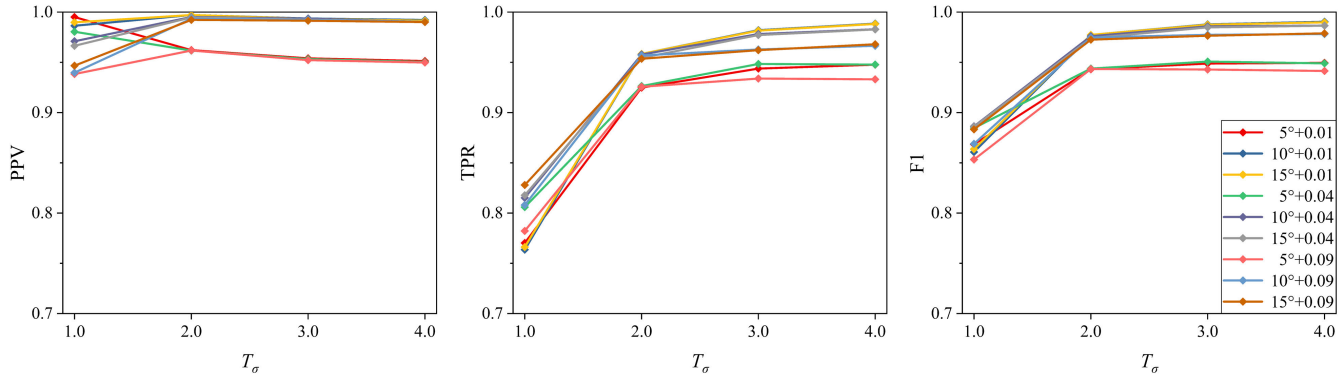


Fig. 8. PPV, TPR, and $F1$ of the proposed method with different parameter settings.

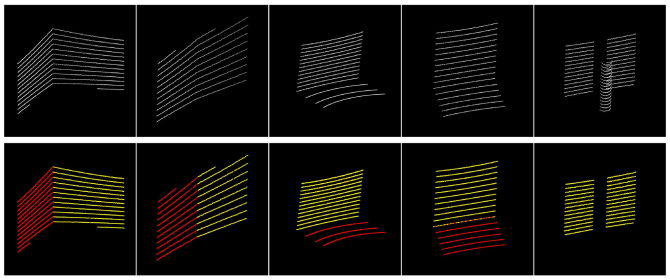


Fig. 9. Examples of the segmentation results on the simulated dataset.

TABLE II
EVALUATION RESULTS ON THE SIMULATED DATASET

	Horizontal	Vertical	Occluded	Average
PPV	0.9974	0.9842	1	0.9935
TPR	0.9783	0.9622	0.9999	0.9782
F1	0.9877	0.9731	1	0.9858

PPV and TPR. In conclusion, considering the processing time and the performance, we choose $T_\sigma = 3.0$, $T_\alpha = 10°$, and $T_e = 0.04$ for this method.

Some examples of the segmentation results are illustrated in Fig. 9. It can be seen that the proposed method is able to distinguish different planes even when a large dihedral angle or occlusion is present. The quantitative evaluation results are given in Table II, and the proposed method achieves an excellent performance with all metrics above 96%. Most points on planar surfaces are extracted and assigned to the correct planes. But some points near the intersection are confused, because we simply assign the nonflat points by comparing distances in line segmentation. The performance in the vertical dataset is impaired by undersegmentation that occurs when the dihedral angle is 170°. Nonetheless, such extreme cases rarely happen in practical application, and thus would make a minor impact on this method.

B. Real Dataset

In Section III-A, we have revealed the effectiveness of our method using the simulated dataset. To further evaluate the proposed method in complex scenarios, five point clouds collected by different types of 3-D LiDAR sensors are used (Fig. 10). We manually created the ground truth data using the segment tool in CloudCompare, a free point cloud processing software (www.cloudcompare.org). As we focus on plane extraction, the rough ground in outdoor scenes was removed. Some basic information of the real dataset is given in Table III, and every point cloud is collected under the rotation rate of 10 Hz.

1) Comparison Baselines and Additional Metrics: Four GPU-free plane segmentation methods are employed for a fair comparison: the NDT cell based RANSAC (NDT-RANSAC) [11], the hybrid region growing (HRG) [26], the PEAC [36], and the line segment grouping (LSG) [30]. NDT-RANSAC is a model fitting-based method designed for unorganized point clouds, whereas the rest of compared methods are designed for organized point clouds. Deep learning-based methods are beyond comparison because they all rely on GPU acceleration. Since the 3-D LiDAR point clouds are sparse and inhomogeneous, we set the cell size in NDT-RANSAC to 1.0 m, which is twice the value used by the authors. Based on the same reason, the size of the subwindow in HRG and the node in PEAC is set to 5×2, i.e., five points in the horizontal direction and two points in the vertical direction, by trial and error. For other parameters in these methods, we adopt the values provided by the authors. The planes with less than 30 points are removed from the results, and thus are not considered during evaluation.

The point-level metrics in Section III-A function well when the planes have similar sizes, but are not adequate for comprehensive evaluation on real-life data. In practice, the planes scanned by the sensor can dramatically vary in size, and point-level metrics are insensitive to the bad performances on those small planes. Thereby, we further add five segmentation metrics: correct detection, oversegmentation, undersegmentation, missed, and noise [59]. Concretely, a mapping process is conducted between the extracted planes and the ground truth. The one-to-one mapping is classified as correct detection. Otherwise, the one-to-multiple mapping is classified as oversegmentation or undersegmentation. For the planes that are not involved in any mapping, those in the ground truth are classified as missed, while those in the extracted planes are classified as noise.

2) Evaluation on VLP-16 Data: We collected point clouds in two scenes using a Velodyne VLP-16 LiDAR sensor, including an underground garage and an urban street (Fig. 11). The

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TABLE III
DESCRIPTION OF THE REAL DATASET

Scene	Bounding Box (m^3)	Average Density ($points/m^2$)	# of Points	# of Planes
1	$94.0 \times 38.0 \times 6.8$	194.67	27454	33
2	$222.7 \times 128.8 \times 32.9$	7.13	15575	28
3	$202.9 \times 104.0 \times 17.2$	260.92	44488	52
4	$192.4 \times 126.3 \times 21.4$	36.74	23660	55
5	$85.5 \times 189.4 \times 20.4$	107.98	31257	23

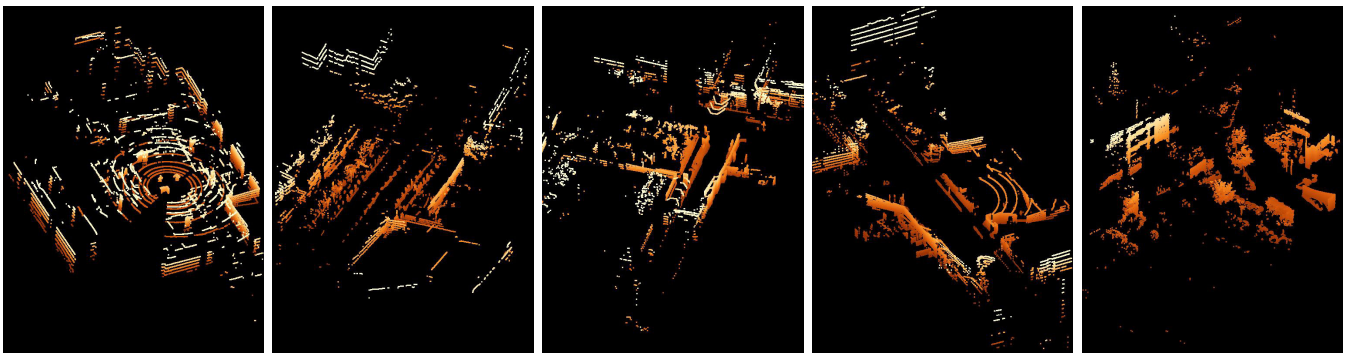


Fig. 10. Input point clouds of the test scenes. From left to right: Scene 1–5.



Fig. 11. RGB photographs of (a) Scene 1 and (b) Scene 2.

planes in Scene 1 can be categorized as horizontal or vertical. The horizontal planes mainly include the ground and the ceiling, while the vertical planes come from walls and square pillars. The proposed method successfully detects most vertical planes, and even the two walls partially blocked by pillars are properly extracted. As for horizontal planes, the ground is correctly detected as one plane. But the ceiling is split into three separate pieces due to the pipes, which is the only oversegmentation (Fig. 12). The NDT cells exploited by NDT-RANSAC are not suitable for the 3-D LiDAR point clouds. Due to the sparse property, the points inside each cell often cannot represent the local surface, and thus NDT-RANSAC yields many noise planes. HRG and LSG correctly extract several vertical planes, but both of them struggle to handle occlusion. Compared to the messy results of NDT-RANSAC and HRG, PEAC produces a tidy result composed of planar surfaces except a wrongly detected pedestrian. However, many planes are missed due to the low vertical resolution of the 3-D LiDAR sensor. PEAC relies on a graph for conducting clustering, where the edges are established by checking three consecutive nodes in the horizontal direction or the vertical direction. For our parameter setting, this means only planes scanned by at least six lasers can be detected. As a result, it is difficult for PEAC to extract distant planes and small planes. On the other hand, PEAC discards the nodes with large differences in range. This can not only avoid spurious planes caused by occlusion, but also lead to miss detection of the planes roughly parallel with the laser, like the ground in Scene 1.

The planes in Scene 2 are mainly from building facades. Since the spatial resolution of the 3-D LiDAR sensor degenerates with distance, the average density of Scene 2 is much lower than that of Scene 1. Suffered from the low density, NDT-RANSAC cannot recover the dominant planar structure, and PEAC fails to detect planes in Scene 2. HRG is less affected but extracts several spurious planes. The reason is that HRG does not impose constraints to avoid the subwindows containing points from disconnected surfaces. For LSG, the trees and poles impede the line segment extraction in the horizontal direction, and thus some planes are not detected. Our method extracts a variety of planar surfaces with different distances, and is robust to the density changes (Fig. 13). This is because we have taken the static angular resolution into consideration when determining parameters.

The quantitative results are listed in Table IV, and we use LBDP to denote the proposed method for brevity. LBDP outperforms the compared methods in terms of PPV, TPR, and $F1$, and enjoys the most correctly detected planes. However, some planes are missed, there are two principal reasons for this: 1) during line segmentation, we discard the segments containing less than three points, so the planes that are only scanned by two lasers cannot be detected and 2) after merging, some narrow planes with $e_2/e_3 \leq T_e$ are mistakenly regarded as cylinders and removed.

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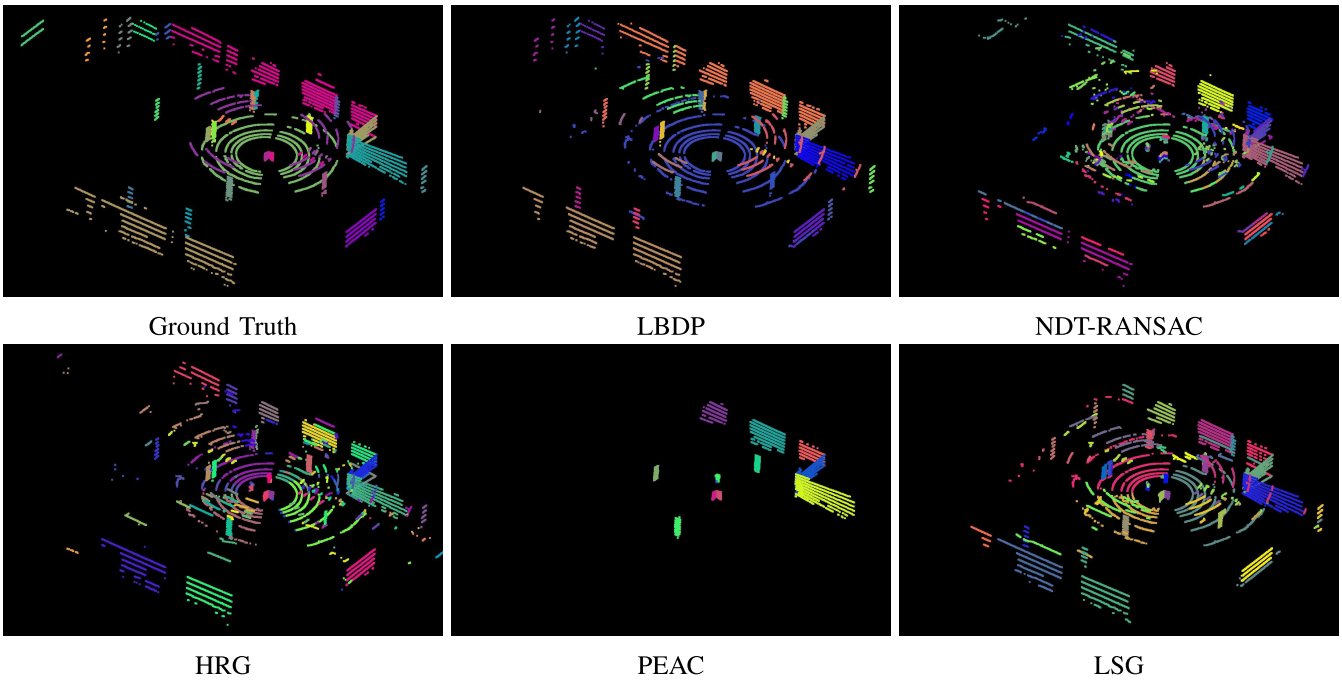


Fig. 12. Qualitative comparison in Scene 1.

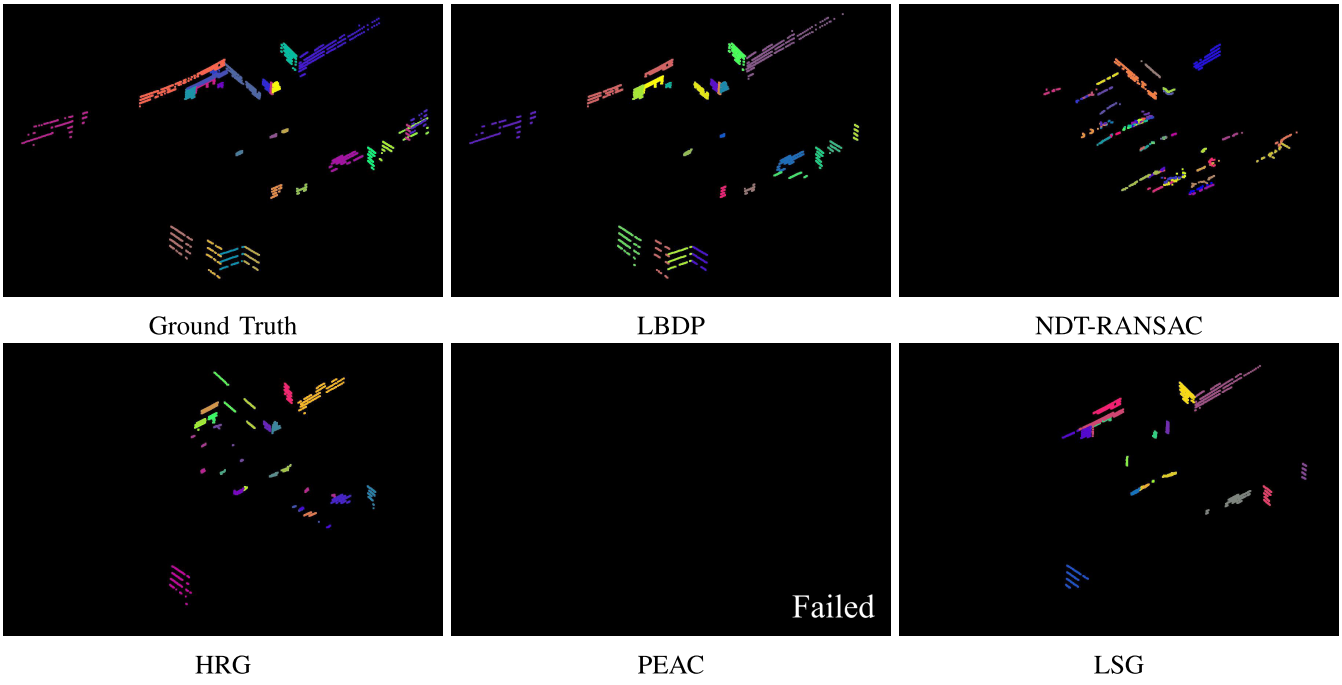


Fig. 13. Qualitative comparison in Scene 2.

3) Evaluation on HDL-32E Data: An open-sourced dataset, the UrbanNav dataset, is employed to show the performance of our method on a different 3-D LiDAR sensor [60]. This dataset was collected using a Velodyne HDL-32E LiDAR sensor, and we select two typical urban scenes for evaluation. The technical parameters of this sensor are given in Table I. Benefitting from more lasers and higher resolution, HDL-32E captured abundant planes from various human-made objects. Meanwhile, rough surfaces that are locally planar are more likely to be extracted. The visual results of different methods on HDL-32E data are shown in Figs. 14 and 15. The hedges in Scene 3 and Scene 4 cause many noise planes in the results of NDT-RANSAC, HRG, and LSG, and also slightly affect LBDP. These rough surfaces are not seen in the results of PEAC, but PEAC only detects a few planes. Besides, HRG extracts noise planes from poles and pillars with rounded corners. By contrast, LBDP splits the cylindrical surfaces into a series of narrow planes and filters out them. As shown in Fig. 16, LBDP correctly detects the planar surfaces of these pillars.

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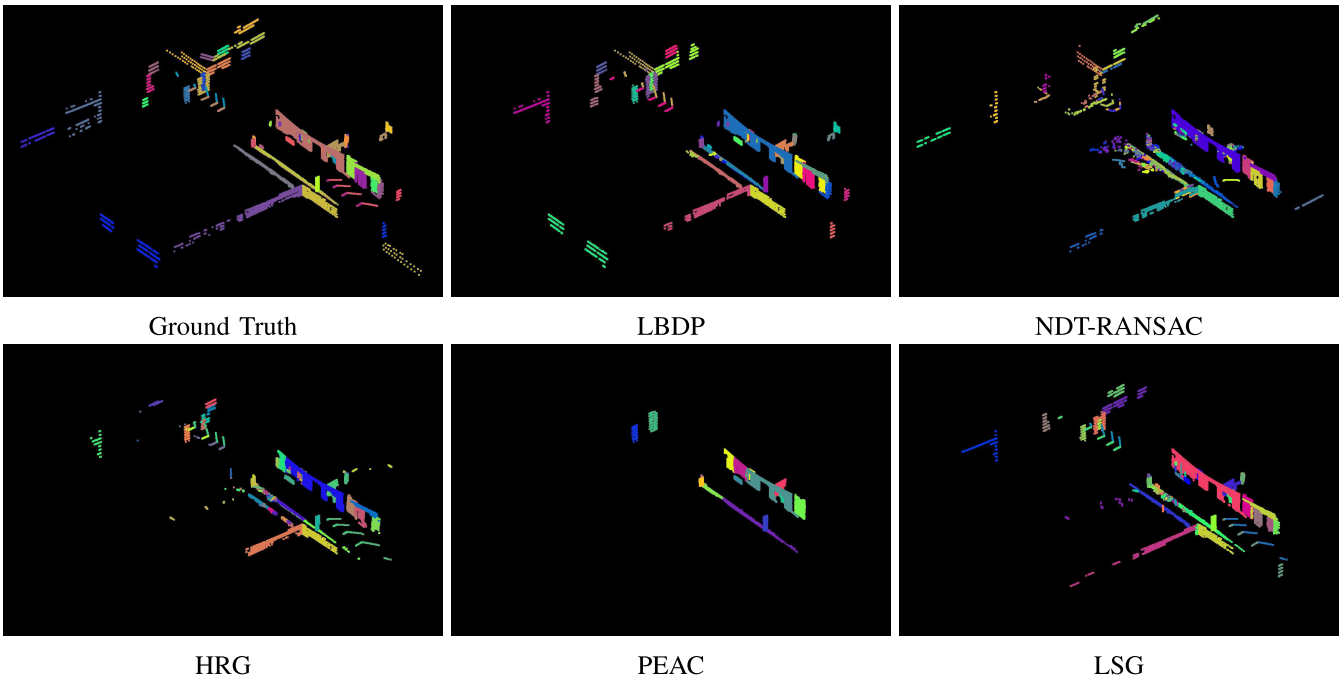


Fig. 14. Qualitative comparison in Scene 3.

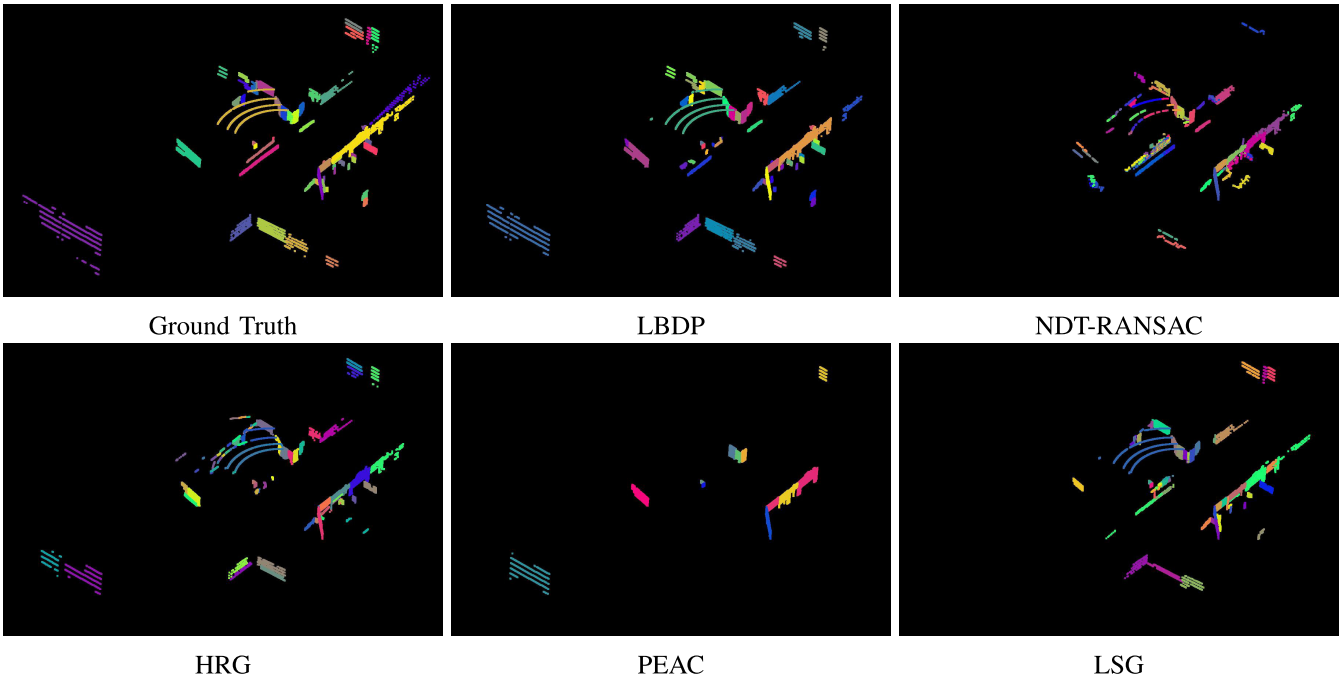


Fig. 15. Qualitative comparison in Scene 4.

Although the average PPV, TPR, and $F1$ of the compared methods are improved in the HDL-32E point clouds, those of LBDP are still the highest (Table IV). PEAC enjoys the least oversegmentation and noise, but many planes are missed and the correct detection is much less than LBDP. Note that we adopt a greedy strategy in merging, and thus some coplanar but disconnected planes are wrongly detected as one plane. Consequently, more undersegmentation can be found in our results. Point cloud registration using planar features could benefit from this property because less planes need to be considered during data association. Nonetheless, the undersegmentation is adverse to 3-D reconstruction.

To demonstrate the efficiency of LBDP, we selected 480 frames out of the UrbanNav dataset with a fixed interval. Generally, the rotation rate of 3-D LiDAR sensors is set to 10 Hz, so the processing time of each frame should be less than 0.1 s to fulfill online computing. A laptop with an Intel i5-9400 CPU of 2.9 GHz is used for the evaluation. Since NDT-RANSAC takes around 1.0 s to segment each fame, we only employ HRG, PEAC, and LSG for comparison.

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TABLE IV
EVALUATION RESULTS ON THE REAL DATASET

Method	Correct detection	Over-segmentation	Under-segmentation	Missed	Noise	PPV	TPR	F1
Scene 1								
LBDP	**26**	1	0	6	1	**0.8815**	**0.8612**	**0.8713**
NDT-RANSAC	2	6	0	25	77	0.4825	0.5281	0.5043
HRG	15	3	0	15	56	0.4115	0.4338	0.4224
PEAC	7	1	0	25	2	0.7080	0.2543	0.3742
LSG	20	5	0	8	12	0.4640	0.5306	0.4951
Scene 2								
LBDP	**23**	0	0	5	1	**0.9440**	**0.7579**	**0.8408**
NDT-RANSAC	1	0	0	27	51	0.1522	0.1306	0.1406
HRG	7	2	0	20	17	0.6439	0.3677	0.4681
PEAC	-	-	-	-	-	-	-	-
LSG	11	0	0	17	6	0.7138	0.5294	0.6079
Scene 3								
LBDP	**36**	2	3	8	6	**0.8942**	**0.8964**	**0.8953**
NDT-RANSAC	12	3	1	35	94	0.5817	0.7177	0.6426
HRG	22	5	0	25	40	0.6772	0.6434	0.6599
PEAC	11	1	0	40	3	0.5708	0.4521	0.5046
LSG	31	2	1	17	43	0.6465	0.6910	0.6680
Scene 4								
LBDP	**43**	2	1	8	9	**0.9250**	**0.8363**	**0.8784**
NDT-RANSAC	12	4	0	39	62	0.4130	0.3957	0.4041
HRG	23	6	0	26	25	0.6240	0.4760	0.5400
PEAC	10	1	0	44	1	0.7176	0.3165	0.4392
LSG	28	5	0	22	17	0.6600	0.5345	0.5907
Scene 5								
LBDP	**19**	0	1	2	4	**0.9641**	0.7699	**0.8561**
NDT-RANSAC	2	1	0	20	11	0.8902	0.5149	0.6524
HRG	8	0	0	15	10	0.9107	0.7064	0.7956
PEAC	4	0	0	19	1	0.9023	0.7104	0.7949
LSG	9	0	0	14	19	0.8914	**0.7953**	0.8406

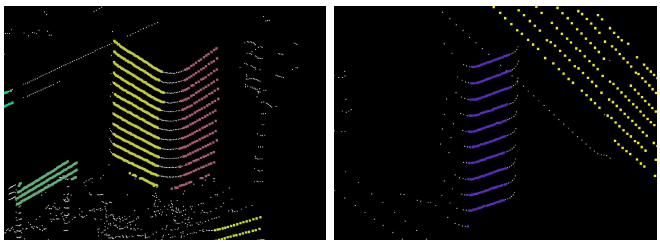


Fig. 16. Segmentation results of LBDP on pillars with rounded corners.

As shown in Fig. 17, the running time of PEAC changes violently and sometimes is beyond 0.1 s. For HDL-32E data, LBDP, HRG, and LSG can all meet the requirement on running speed, with the average process time of 15.6, 6.2, and 26.2 ms, respectively. HRG is about 2.5× faster than ours because subwindows are directly obtained from the image structure whereas line segments are not. On the other hand, the running time of LBDP depends on the number of line segments, and thus fluctuates in different scenes. However, the maximum is 22.5 ms in this test, and does not prevent LBDP from online processing.

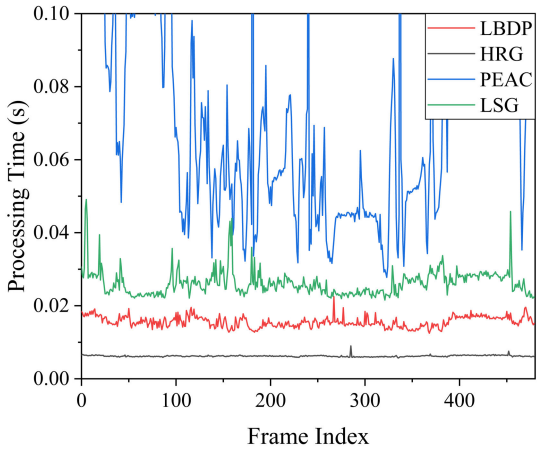


Fig. 17. Time performance of LBDP, HRG, PEAC, and LSG on the UrbanNav dataset.

4) Evaluation on OS1-64 Data: We also test the proposed method on the point cloud composed of 64 scan lines, using the MulRan dataset [61]. An Ouster OS1-64 LiDAR sensor was utilized in this dataset, with only 290° field of view

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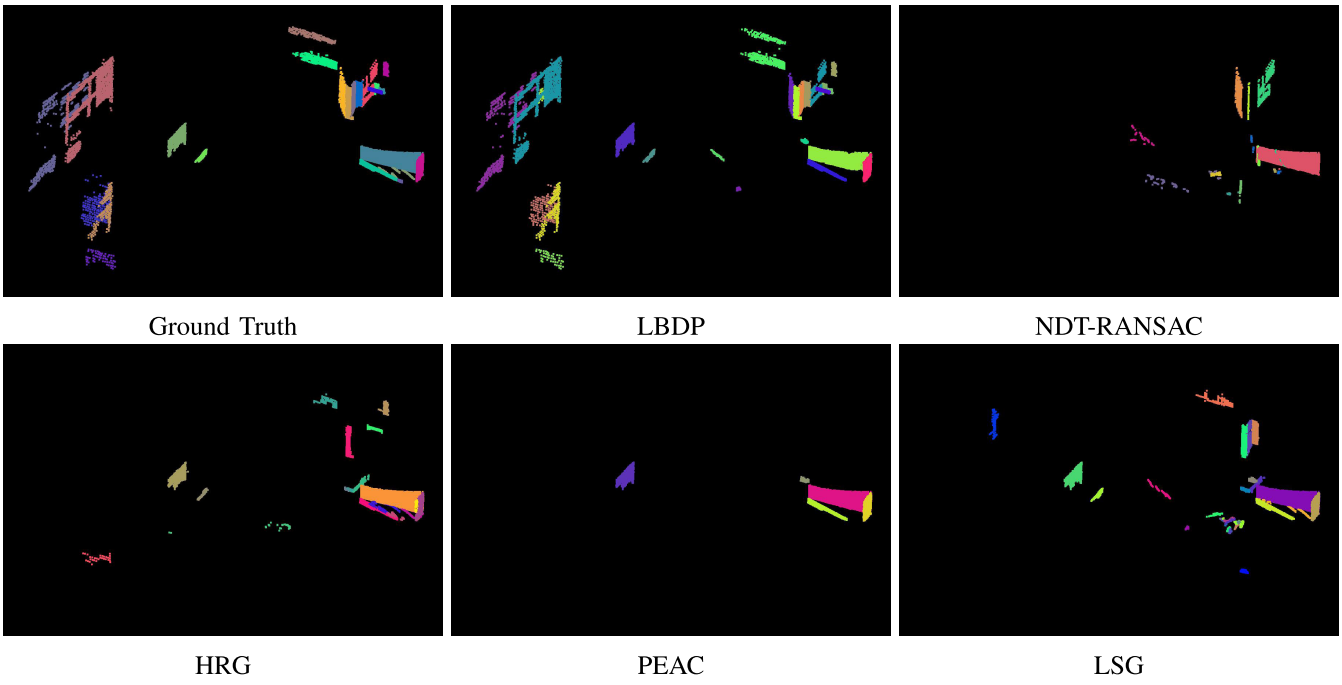


Fig. 18. Qualitative comparison in Scene 5.

available due to the radar behind it. Scene 5 was captured in the KAIST campus. LBDP extracts 19 out of 23 labeled planes, outperforming other methods. But, the result contains a few noise planes from parking cars (Fig. 18). These parking cars can also be found in the results of baseline methods. Besides, LSG is prone to take rough surfaces as planes, and it extracts 11 spurious planes from the trees. Although LBDP shows superior performance in terms of segmentation metrics, LBDP and LSG are comparable with regard to point-level metrics. The largest plane accounts for more than half of the points in the ground truth. However, 72% points on this plane are correctly assigned by LBDP, which lowers the recall. The possible reason is that the noise model parameters are not specifically calibrated for ouster sensors.

So far, we have tested the proposed method on diverse datasets. Despite the LiDAR sensors with different angular resolution and field of view, the collected point clouds are always organized and can be projected onto an image. As a consequence, the results show that our method is applicable to various 3-D LiDAR sensors.

IV. Conclusion

This article proposes a novel point-to-line-to-plane framework that can efficiently extract planar surfaces from sparse 3-D LiDAR point clouds. We first group discrete points into line segments by investigating their flatness. Instead of using region growing, which suffers from the poor geometry of the selected initial region, we then present the LBDP algorithm to detect planes from line segments. Finally, region growing and merging are employed to gain more complete results. Moreover, to reduce the use of empirical thresholds and improve the robustness, we exploit the predefined noise model to adaptively determine reasonable thresholds throughout the proposed method. The approach is tested on both simulated dataset and real dataset. With fixed parameters, our method shows consistent and robust performance on the real dataset, which involves different devices and scenes. Furthermore, the proposed method outperforms the compared methods, especially when dealing with varying point density and occlusion. Benefitting from the line segmentation, the mean runtime of our method is less than 20 ms/frame and thus qualified for real-time application.

As fundamental geometric features, the extracted planes can be applied to many point cloud processing steps, such as calibration, registration, and modeling. More specifically, 3-D LiDAR sensors are usually mounted on intelligent vehicles and robots, so the proposed method is expected to contribute to LiDAR odometry and SLAM techniques. At present, we merely use a rough model to describe the statistics of measurement noise, which plays an important role in our method. In the future, a more accurate model considering additional factors, such as range and intensity, would help produce more robust and reliable results.

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